



FOR JAVA DEVELOPERS MOVING TO NESTJS

Debit vanished.

Spring Boot to NestJS, episode 16: MongoDB with Mongoose

Changing frameworks did not remove the race. Validation belongs to the write path, not the field.

Two requests loaded the same account and each saved its own debit. The balance ended at 50 instead of 20, on Spring Data MongoDB and on Mongoose alike, with no error.

Detect stale state with `@Version` or `optimisticConcurrency`, or avoid the read-modify-write with an atomic `$inc`.

Validation depends on the write path on both stacks, and `$inc` walks past Mongoose update validators. Put the rule in the filter.

In scope

- `@Document` and `MongoRepository` against Schema and Model
- Lost updates and optimistic concurrency
- Validation by write path
- Casting and strict mode

Not in scope

- Multi-document transactions
- `populate` against `DBRef`

One machine, one account, the same MongoDB for both stacks.

01 The measurements

TWO LOADS, TWO SAVES	balance 50 on both stacks, expected 20
@Version / optimisticConcurrency	second save rejected, balance 70
ATOMIC \$inc	balance 20 on both stacks
SPRING save(), @Min(0)	saved -500; rejected with ValidatingEntityCallback
MONGOOSE save(), min: 0	ValidationError
MONGOOSE updateOne()	stored -500; rejected with runValidators
DIRECT UPDATE, BOTH STACKS	stored
\$inc -500 WITH runValidators	stored, balance -400
\$inc WITH balance >= 500 IN THE FILTER	matched 0, balance 100

Balance 100, debits of 30 and 50, and a rule of minimum 0.

02 Detect it, or avoid it

Optimistic concurrency keeps the read-modify-write and rejects stale state. An atomic update removes the read entirely.

```
probes.ts
```

```
new Schema(accountShape, { optimisticConcurrency: true })

Account.updateOne({ _id }, { $inc: { balance: -30 } })
```

```
Ep15GuardedAccount.java
```

```
@Version
private Long version;
```

two concurrent debits of 30 and 50 from 100: 50 unguarded, 70 plus an error with a version check, 20 with \$inc

Run on a real machine, 2026-09-30, by scripts/verify-mongo.sh in the course repository.

03 Put the rule in the filter

Mongoose update validators only run on \$set, \$unset, \$push, \$addToSet, \$pull and \$pullAll. For \$inc, make the rule part of the match.

```
probes.ts
```

```
await Account.updateOne(
  { _id, balance: { $gte: 500 } },
  { $inc: { balance: -500 } },
);
```

\$inc -500 on 100 with runValidators: -400; with the rule in the filter: matched 0, balance 100

Run on a real machine, 2026-09-30, by scripts/verify-mongo.sh in the course repository.

Before you ship Mongoose where you had Spring Data MongoDB

1. Turn on optimisticConcurrency where you read, change and save
2. Prefer one atomic update when the change allows it
3. Pass runValidators on updates, and know it skips \$inc
4. Put numeric rules for \$inc in the filter
5. Remember strict mode drops undeclared fields silently
6. In Spring, register ValidatingEntityCallback if you rely on @Min

Everything else

 Site https://code-with-sam-dev.github.io	 Facebook https://www.facebook.com/codewithsamdev
 YouTube https://www.youtube.com/@CodewithSam-Dev	 GitHub https://github.com/code-with-sam-dev
 LinkedIn https://www.linkedin.com/company/code-with-sam-dev	 Buy me a coffee https://buymeacoffee.com/codewithsam
 x https://x.com/CodeWithSamDev	 Buy me sadza https://buymesadza.com/codewithsam
 TikTok https://www.tiktok.com/@codewithsamdev	Every measurement, one command: scripts/verify-mongo.sh https://github.com/code-with-sam-dev/spring-to-node
 Instagram https://www.instagram.com/codewithsamdev	Mongoose validation https://mongoosejs.com/docs/validation.html

If this was useful, a like and a subscribe help more than you would think. And tell me in the comments which part of your workflow you want taken apart next.

Spring is a trademark of Broadcom. Java is a trademark of Oracle. Node.js is a trademark of the OpenJS Foundation. MongoDB is a trademark of MongoDB, Inc. This is an independent, unofficial guide produced by Code with Sam, not affiliated with or endorsed by any of them.

Technical claims verified 2026-09-30 against the documentation named beside each one. Vendors change behaviour and defaults between versions, so check the version you are actually running before you rely on any of it.